

Midhilesh Momidi

Bangalore • midhileshmomidi1790@gmail.com • 9347971191

EXPERIENCE

Staff ML Engineer @ WALMART

Nov 2024 – Present

- Developed a **Multi-Agentic RAG** architecture using **LangGraph** with **Redis** for rapid agent-state retrieval and real-time voice-agent Speech to Text (**STT**) replacing existing **IVR Call Routing**
- Achieved **4 - 7 sec** Time-to-First-Token (**TTFT**) for 80% calls caching FAQ responses and eliminated redundant handoffs to human agents, delivering an estimated **200,000+** hours in annual operational savings
- Developed a real time **monitoring** and observability loop using **Kafka** to store the telemetry data to **Azure Cosmos DB** and **Big Query**
- **Fine-tuned** a Policy Recommender model using the **Llama-2 13B** model with Parameter-Efficient Fine-Tuning (**LoRA**), serving **2.3M** associates, and optimized inference throughput using **vLLM**
- Developed and deployed a scalable Translation App for Walmart store associates with a daily active usage of 20,000+ users leveraging Azure Cognitive Services for **real-time** Text-to-Speech (**TTS**), Speech-to-Text (**STT**), and Text-to-Text (**TTT**) translations utilizing Kafka and **WebSockets**

Senior ML Engineer @ DELL TECHNOLOGIES

OCT 2021-Nov 2024

RAG Pipeline for Dell Chat Assistant

- Led a team of 4 engineers to develop Retrieval-Augmented Generation (**RAG**) pipeline to production serving 2000 requests per day with end-to-end software architecture ensuring model reliability across different nodes
- Utilized **Postgres Vector DB** for efficient storage and retrieval of embeddings using proper indexing like **IVF** and **HSNW** thus enhancing relevant content generation using open-source Large Language Models (**LLMs**) like Llama2-13B serving with **vLLM** for faster inference
- Use **Ragas** for evaluating the retrieval and generation metrics for efficient quality of the responses to the user and monitor the metrics through Grafana and develop a feedback loop using **Kafka** to store the responses and metrics to Big Query for analytics
- Incorporated a **Reflexion** Design Pattern mechanism where outputs are iteratively refined reducing hallucinations and improving factual accuracy
- Utilized **Celery** for distributed task queue management, with **RabbitMQ** as message queue and Postgres as the backend, to streamline complex data preprocessing operations such as chunking, document and html parsing, embedding generation, enhancing efficiency in managing embedding uploads to Postgres Vector db
- Reduced the latency from **7-8 sec** to **3-4 sec** utilizing model optimization and quantization and adding **tags** in VectorDB for the documents

Cross Sell Recommendations

- Developed an end-to-end machine learning model for **personalized** recommendations, effectively for 30 million users across US, UK, Canada, APJ suggesting 5-8 relevant Cross sell and Upsell recommendations serving under **400ms** latency in peak time
- Developed embeddings for Order Codes and SKUs (peripherals) using multi stage retrieval strategy utilizing **Two Tower Network** architecture utilizing **Tensorflow** and **Transformer** architectures, leveraging multi-GPU training to accelerate training on large datasets
- Developed data pipelines for retrieving data from Hadoop and Greenplum, using **PySpark** and **Airflow** for scheduling daily data synchronization tasks and update the data to **Feature Store** to both **online** and **offline** tables for ranking model
- Leveraged Redis as an in-memory database for low-latency retrieval of real-time predictions, and designed fallback strategies (popularity-based, item-based) to handle cold-start scenarios for new users and products
- Built and deployed a **FastAPI** service integrated with Redis and MongoDB for scalable recommendation retrieval, orchestrated on Kubernetes to ensure fault tolerance, efficient scaling, and high availability

Predictive Models for Next Best Action and Visit Intent and Demand Modeling

- Implemented sequence-based deep learning models using **LSTM** to predict the next best action for product brand categories, leveraging multi-GPU training using **Pytorch** to handle large-scale data efficiently, enhancing UK laptop sales by 16% through targeted brand recommendations
- Utilized **AWS Glue** crawlers to manage and discover data stored in S3, enabling seamless integration with downstream machine learning workflows
- Applied Double Machine Learning (**Double ML**) techniques to accurately predict baseline product **prices**, achieving optimal **price elasticity** of demand
- Developed a multiclass classification model using **XGBoost** to discern user intent—whether visitors to the site came to shop, browse, or seek support—enhancing user experience and site engagement
- Leveraged **Sagemaker** endpoints to serve batch predictions, allowing for scalable and efficient deployment of machine learning models

Sr. Data Scientist @ ERNST & YOUNG

MAR 2021-Sep 2021

- Developed an ETL pipeline using PySpark to effectively process millions of rows fetching data from different resources like extracting data from invoices using Camelot, Greenplum, Hadoop etc and Airflow as the scheduler to schedule the execution of Spark jobs
- Developed a customizable analytics dashboard using **Plotly** and integrated it into a backend web application using Django, enhancing data visualization and user interaction
- Created a data pipeline to extract invoice information from PDFs using Camelot, and subsequently developed a web application using Django to provide a user-friendly interface for accessing and managing the extracted data

Data Scientist @ TCS

MAR 2016-Feb 2021

- Developed a topic modeling algorithm using Latent Dirichlet Allocation (**LDA**) to categorize 900 million in-house IT tickets. Implemented with distributed ML library **Spark MLlib**, this system efficiently routes tickets to the appropriate agents, improving service level agreement (SLA) response times from 3 days to 1 day
- Engineered neural network and tree-based models to predict flight arrivals within a +/- 5-minute window, enhancing operational accuracy and traveler satisfaction
- Built a data pipeline using **Beautiful Soup** to scrape real-time data from the FlightStats website. Integrated this data into a model pipeline using AWS **SageMaker**, with performance metrics monitored through **CloudWatch**, ensuring continuous model optimization
- Developed classification models to predict room types, hotel categories, discounts, and rewards for users in the hospitality industry, significantly enhancing guest experiences
- Improved model performance from 65% to 75% by incorporating seasonality factors related to different time slots, providing more granular and in-depth analysis for strategic decision-making
- Reduced the latency of I/O operations for reading from S3 buckets by implementing multithreading and caching strategies, streamlining data access and processing

EDUCATION

JNTUA – B. TECH [GPA: 9.1]	2011-2015
Sri Chaitanya Jr. College - 12th [GPA: 10]	2009-2011
Sai Vidyanikethan EM School - SSC [GPA:9.8]	2009

SKILLS

Python, Machine Learning, Deep Learning, Pytorch, Tensorflow, NLP, Pyspark AWS, Sagemaker, Dockers, Kubernetes, CICD, Airflow, MLFlow, FastAPI, Flask, MongoDB, Recommender Systems, Spark ML, Hive, Hadoop, AWS, Transformers, NoSQL, OOPS, LLMs, MLOps, Langchain, GPT, Llama, Model Optimization, Quantization, Fine Tuning, LangGraph, Agentic AI/RAG, Cassandra